5.1 Introduction

Electric drive or electric motor is a device which converts electrical energy into mechanical energy to do a particular work connected to it.

The electric drive designed electric train is shown in figure 5.1. It is operated by a high-torque electric DC series motor. The system used to control the speed is called power drive. While using this electric motor, its speed is controlled by the current, voltage and frequency.



Fig. 5.1 Electric train

5.2 Types of electric drives

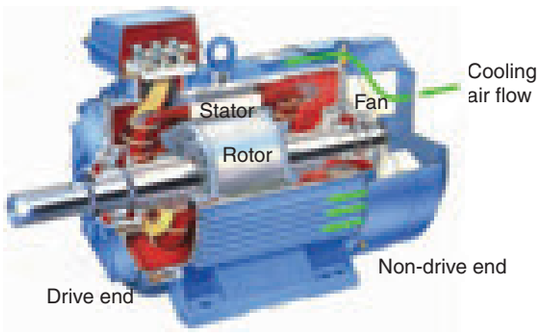


Fig. 5.2 Electric motor

The internal construction of a motor is shown in figure 5.2. Depending on the applications, the motor can be divided into different categories. According to the design, the drives are classified as

i. Individual drive,
ii. Group drive and
iii. Multi-motor drive.

5.2.1 Classification of electrical drives

1. Depending on the power supply,
   A. Direct Current electric drive
   B. Alternating Current electric drive
2. According to the process of electric drives,
   A. Continuous duty drive
   B. Short-term duty drive
   C. Intermediate duty drive
3. According to the control of Drives,
   A. Manual
   B. Semi automatic
   C. Automatic
4. According to the number of machines,
   A. Individual drive
   B. Group drive
   C. Multi-motor drive
5. Based on speed control
   A. Reversible and non reversible speed control
   B. Variable speed control

5.2.2 DC drive

The drive is nothing but one type of electric machine. This converts electrical energy into mechanical energy.Its structure is similar as in figure 5.3. The speed of the drive will depend on the amount of voltage given to it. Small type motor is used in sports equipment such as toys. Large size of electric motor is used in electrical vehicles such as Hoists and Steel rolling mills.

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